

A Packing of 19 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 19 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.345324. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.270289, obtained by deleting 26 of the 45 four-spaces of the packing listed in Cohn’s current table. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 19 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.345324 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 19$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.270289, obtained by deleting 26 of the 45 four-spaces of the packing listed in Cohn’s current table. The Rankin simplex bound for this cell is $4(9-4)/9 \cdot 19/18 = 2.345679$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N19.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
586331007516190931515955332548633425908150208922937294826059214910093309
192976319507319322038642813588605086657700898087585434624086845509264474
846970859163659463986177822655454891376275732665
```

denominator:

```
2499999999999891178125321646529338468455219771357312039288108695622696896
807271887701655783713066875132032522846238150027018386599636652780025714
166093826761952957231329648255533394007374643332
```

B Coordinates

19 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N19.txt`.

```
181525530035 542396390969 -160573155059 393294800857277275418778 154080558780 -437998031843 -289513153422344307541 411321519385 327383545256 -320761533732
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